Time from now:	Event
≳188 billion years	Big Rip
Time before Big Rip:	
2 billion years	Erase galaxy clusters
140 million years	Destroy Milky Way
7 months	Unbind Solar System
1 hour	Earth explodes
10^{-19} seconds	Dissociate atoms

Figure 15: Big Rip timeline (based on current worst-case scenario for w), adapted from Caldwell, Kamionkowski, Weinberg, 2003. The time until the Big Rip is at least about 188 billion years. The table indicates other moments of destruction in terms of approximately how long before the Big Rip they would occur.

Phew.

But given the potential consequences, both for the universe and for the structure of physics itself, we in the astronomical community put a pretty high priority on figuring out where we currently sit on the scale from w = -1 to Violent Cosmic Doom.[fn5] We can't measure w directly, but we can determine it indirectly by measuring the past expansion rate of the universe and comparing it to our best theoretical modeling of what different kinds of dark energy would have done. We glossed over this a bit in the previous chapter, but it turns out that even just determining the past expansion rate is far more difficult than it seems like it has any right to be. In principle, there are several ways to get at w, and some of them can be done in subtle ways that don't require calculating the expansion rate at specific distances. But the most straightforward way to get a handle on dark energy is to figure out our full expansion history. And it turns out all the weirdnesses of cosmology come crashing together if you try to do something as simple as answer the question, "How far away is that galaxy?"

LADDER TO HEAVEN